

Discrete Structure

CSIT — 2nd Semester — 2081 — Answer Sheet
Subject Code: CSC165

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. List any four rules of inferences. Using direct and indirect proof show that, for any real number x , if $x^2 - 7x + 12 = 0$, then $x = 3$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What are the necessary and sufficient conditions for graphs to have Euler circuit? Let R be a relation defined on set of natural numbers N , such that $a, b \in N$, $a R b \iff b = 2^k a$, where $k \in \{0, 1, 2, \dots\}$. Show that R is a partial ordering relation on N .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Determine whether the function $f(x) = x^2 - 1$, from set of integers to the set of integers is injective, surjective or bijective. Write the types of fuzzy set operations with their definitions. Give the floor and ceiling value of 1.2 and -1.2 .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Find the GCD of 12 and 16 using Extended Euclidean Algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Using mathematical induction show that $2 + 5 + 8 + \dots + (3n - 1) = n(3n+1)/2$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. State sum rule and product rule. If 26 integers are chosen from the set of consecutive integers $\{1, 2, 3, \dots, 50\}$, prove that there are sure to be two numbers so that one is multiple of the other.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Discuss about meet and join operation between Boolean matrices. What are the values of $5 \bmod 7$ and $-5 \bmod 7$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define chromatic number. How does Kruskal's algorithm find Minimum Spanning Tree?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Solve the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with initial conditions $a_0 = 2$ and $a_1 = 7$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is network flow? Give an example of saturated edge, unsaturated edge and slack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. When do we use permutation rather than combination? How many 5 digit numbers can be generated using the digits 0 to 9, if each number starts with 98 and no digit appears more than once?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Define subset and power set. How do you prove correctness of recursive algorithm using induction? Illustrate with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.